

JEE MAIN REVISION MASTER: ISOMERISM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 10 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Structural Isomerism: Chain, Position & Functional	7 PYQs	Isomerism
Moderate	Stereoisomerism: Geometrical & Optical Isomerism	3 PYQs	Stereochemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Structural Isomerism: Chain, Position & Functional	$\begin{aligned} \text{Isomer Count: } & \text{C}_4\text{H}_{10} \implies 2, \quad \text{C}_5\text{H}_{12} \implies 3, \quad \text{C}_6\text{H}_{14} \implies 5, \quad \text{C}_7\text{H}_{16} \implies 9 \end{aligned}$
Stereoisomerism: Geometrical & Optical Isomerism	$\text{Number of optical isomers for molecule with } n \text{ asymmetric carbons: } 2^n$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Structural Isomerism: Chain, Position & Functional (Moderate Yield — 7 PYQs)

Core Principle: Structural isomerism: compounds having the same molecular formula but different structural arrangements. Pentane (C_5H_{12}) has 3 isomers (n-pentane, isopentane, neopentane). Hexane (C_6H_{14}) has 5 chain isomers.

■ **Caution / JEE Pitfall:** Functional isomers: Alcohols and Ethers; Aldehydes and Ketones; Carboxylic acids and Esters.

■ **Speed Trick:** Boiling points of chain isomers decrease with increasing branching.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The number of chain isomers possible for the hydrocarbon with molecular formula C_5H_{12} is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Among the following, identify the compound that is not an isomer of hexane..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): "Propanone and propanal are..." [Ans: N/A]

3.2 Stereoisomerism: Geometrical & Optical Isomerism (Moderate Yield — 3 PYQs)

Core Principle: Geometrical isomerism (cis/trans) arises due to restricted rotation about $C=C$ or ring with two different groups on each carbon. Optical isomerism requires non-superimposable mirror images (chiral center with 4 different groups, lack of plane of symmetry).

■ **Caution / JEE Pitfall:** Propene cannot show geometrical isomerism because one of the double-bonded carbons has two identical hydrogens.

■ **Speed Trick:** Meso compounds contain chiral carbons but possess an internal plane of symmetry, making them optically inactive.

Representative Source PYQs:

- [2023 Shift] Q#4 (MCQ): "A pair of compounds having the same boiling points are..." [Ans: N/A]
- [2018 Shift] Q#3 (MCQ): "Identify the following compound which exhibits geometrical isomerism..." [Ans: A]
- [2018 Shift] Q#4 (MCQ): "The two forms of D-glucopyranose are called..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The number of chain isomers possible for the hydrocarbon with molecular formula $C_{12}H_{26}$ is

- (A) 4
- (B) 3
- (C) 2
- (D) 1

Correct Answer: None

Solution Analysis: For the hydrocarbon with molecular formula $C_{12}H_{26}$, we need to count the possible chain isomers. $C_{12}H_{26}$ is an alkane. Chain isomerism occurs when the carbon skeleton is arranged in different ways. The possible structures are: Straight chain: $CH_3-CH_2-CH_2-CH_2-CH_3$ This is pentane. One methyl branch: $CH_3-CH(CH_3)-CH_2-CH_3$ This is 2-methylbutane. Two methyl branches on the middle carbon: $C(CH_3)_4$ This is 2,2-dimethylpropane. So, the

Benchmark Problem 2: KCET 2023 (Q#4)

A pair of compounds having the same boiling points are

- (A) cis but-2-ene and trans but-2-ene
- (B) n-hexane and neo-hexane
- (C) benzene and naphthalene
- (D) (+) butan-2-ol and (-) butan-2-ol

Correct Answer: None

Solution Analysis: The correct answer to the question: "A pair of compounds having the same boiling points are," is Option D: (+) butan-2-ol and (-) butan-2-ol. Let's explain why the other options are incorrect: Option A: cis but-2-ene and trans but-2-ene have different spatial arrangements around the double bond. This difference in geometry leads to different physical properties, especially boiling points. Typically, the trans isomers are less polar and have a lower boiling point compared to the cis isomers. Option B: n-hexane and neo-hexane (more correctly called 2,2-dimethylbutane) are structural isomers,